



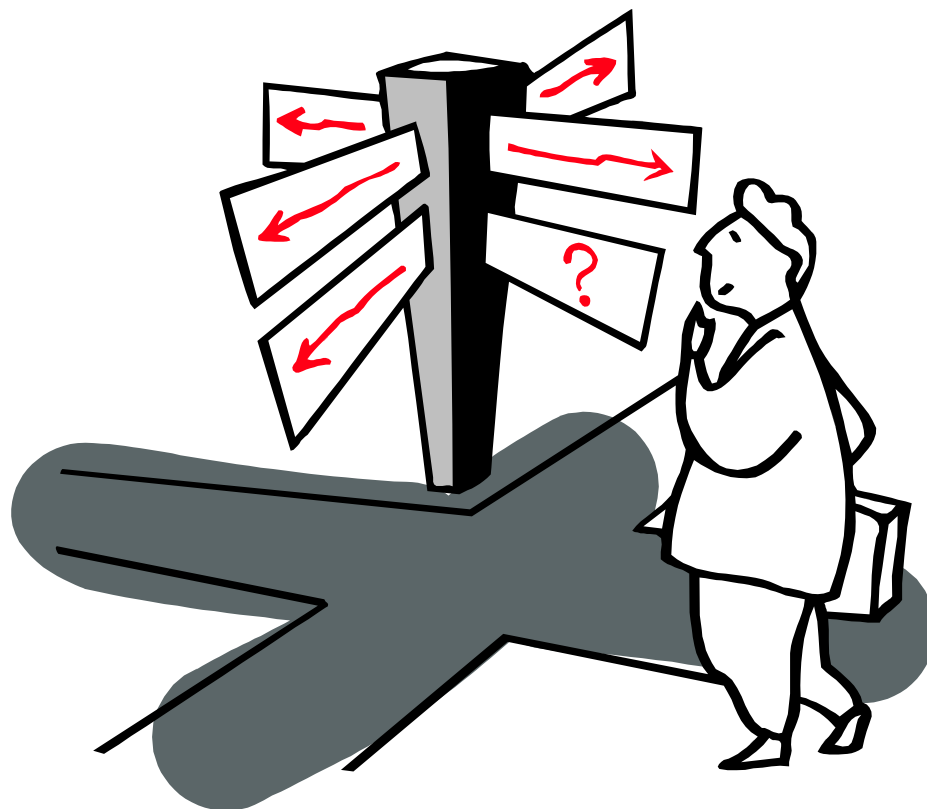
软件体系结构

第5章 仓库与黑板

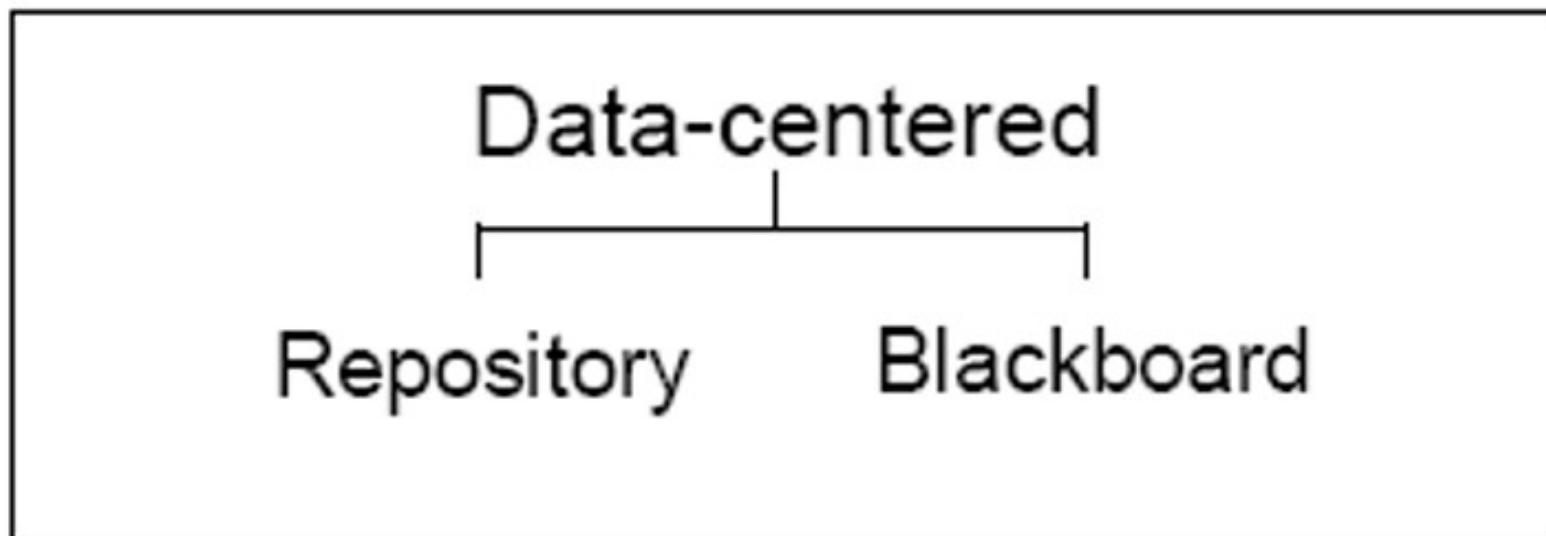
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Families of Architectural Styles



- 基本思想

- 仓库是存储和维护数据的中心场所。
- 仓库式风格的两种组件：中央数据结构组件、相对独立的组件集合

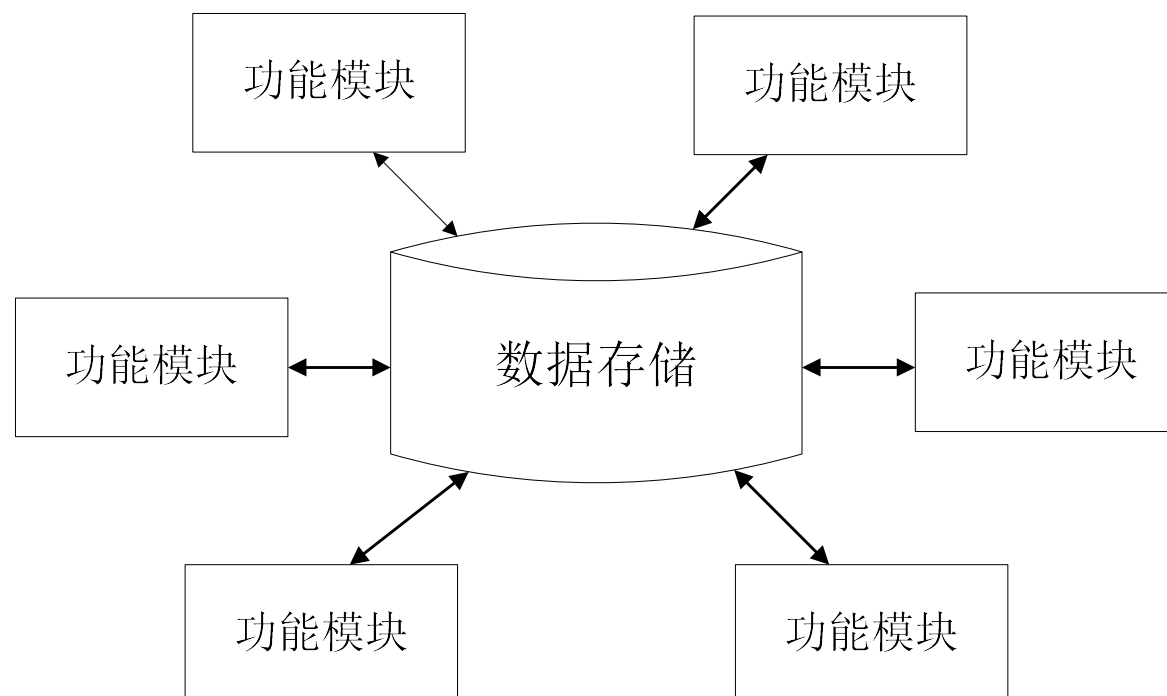


图4.21 仓库风格

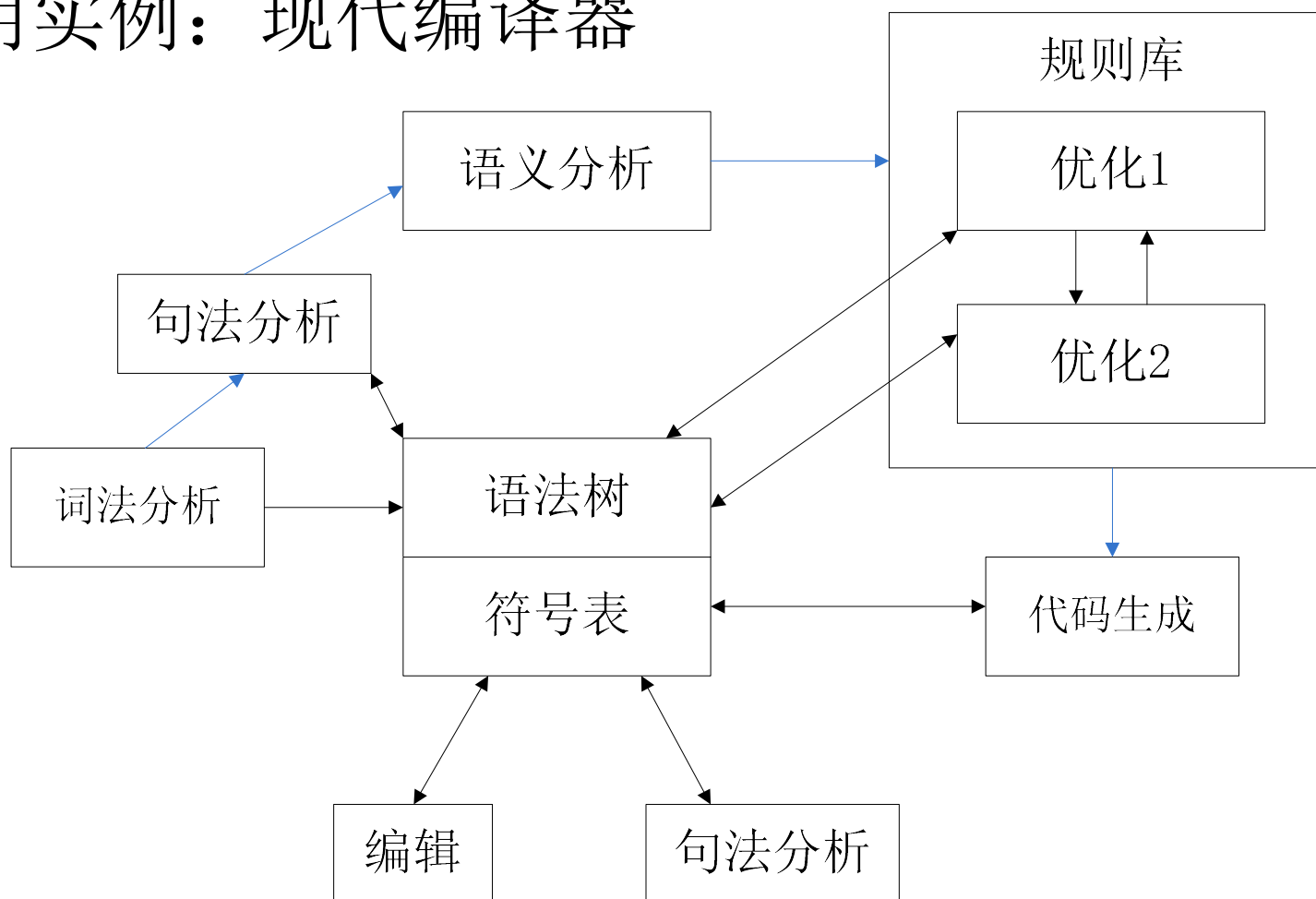
- 优点
 - 便于模块间的数据共享，方便模块的添加、更新和删除，避免了知识源的不必要的重复存储等。
- 缺点
 - 对于各个模块，需要一定的同步/加锁机制保证数据结构的完整性和一致性等。

仓库风格 (3)



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- 应用实例：现代编译器

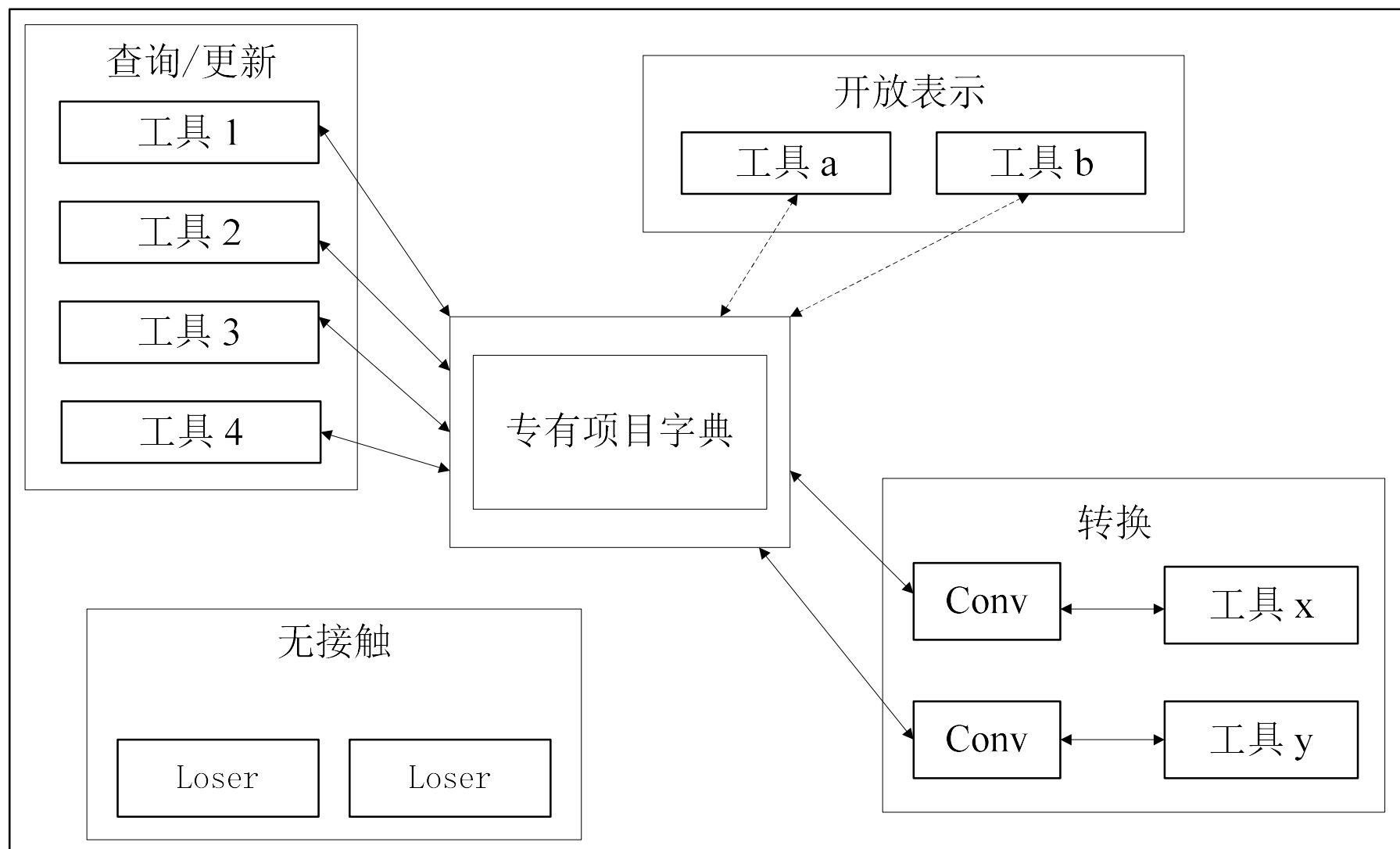


现代的规范编译器结构

仓库风格 (4)



- 应用实例：基于仓库风格的软件研发环境Eclipse



基于仓库风格的Eclipse软件研发环境

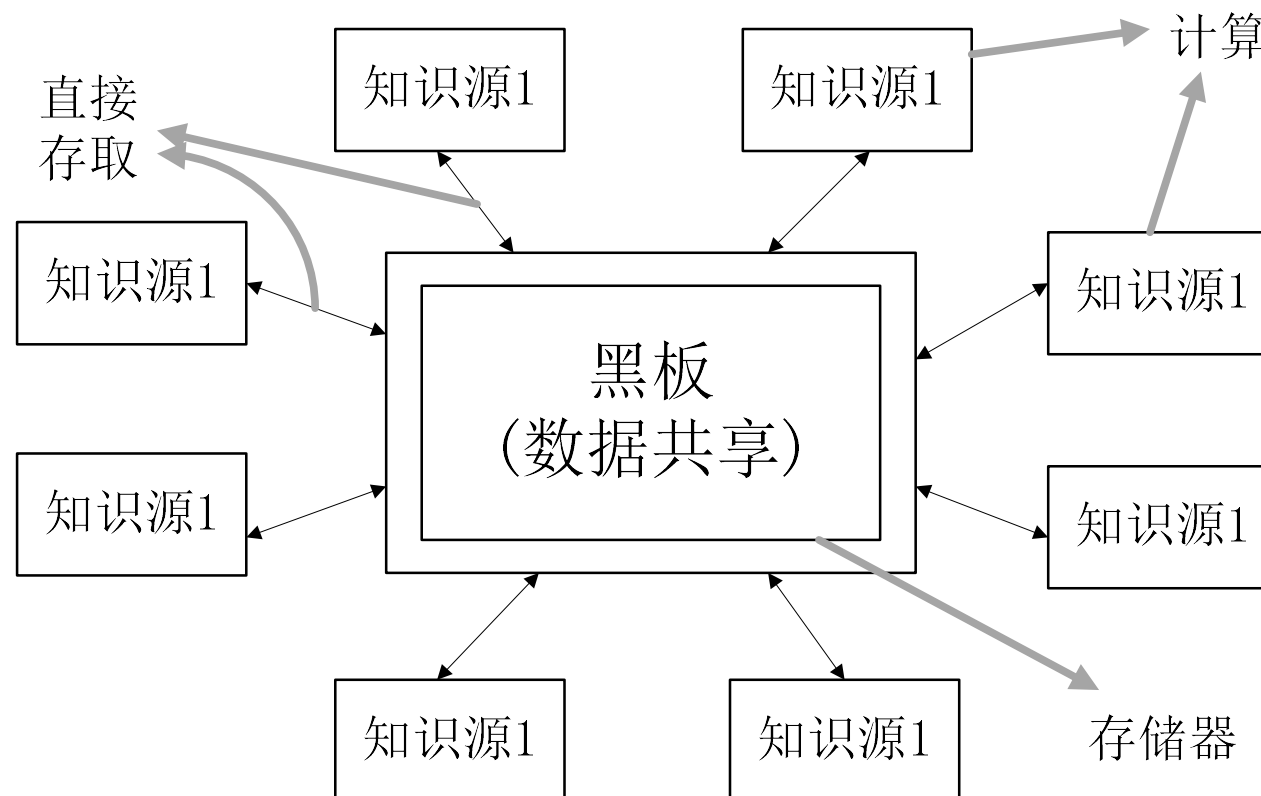


Repositories System

- Style represents a large variety of systems
这种风格描绘很多种系统
 - many variants depending on nature of shared data
共同特点是共享数据
- Style addresses mechanisms for:
风格说明其运行机制是
 - collecting, manipulating and preserving large bodies of data
收集、操作、保存大量的数据
- KB is a natural example.
知识库是一个典型的例子。
 - 具有成长的特性。

Repository (Blackboard)

- 组成部分
 - (1) 知识源 (2) 黑板数据结构 (3) 控制器

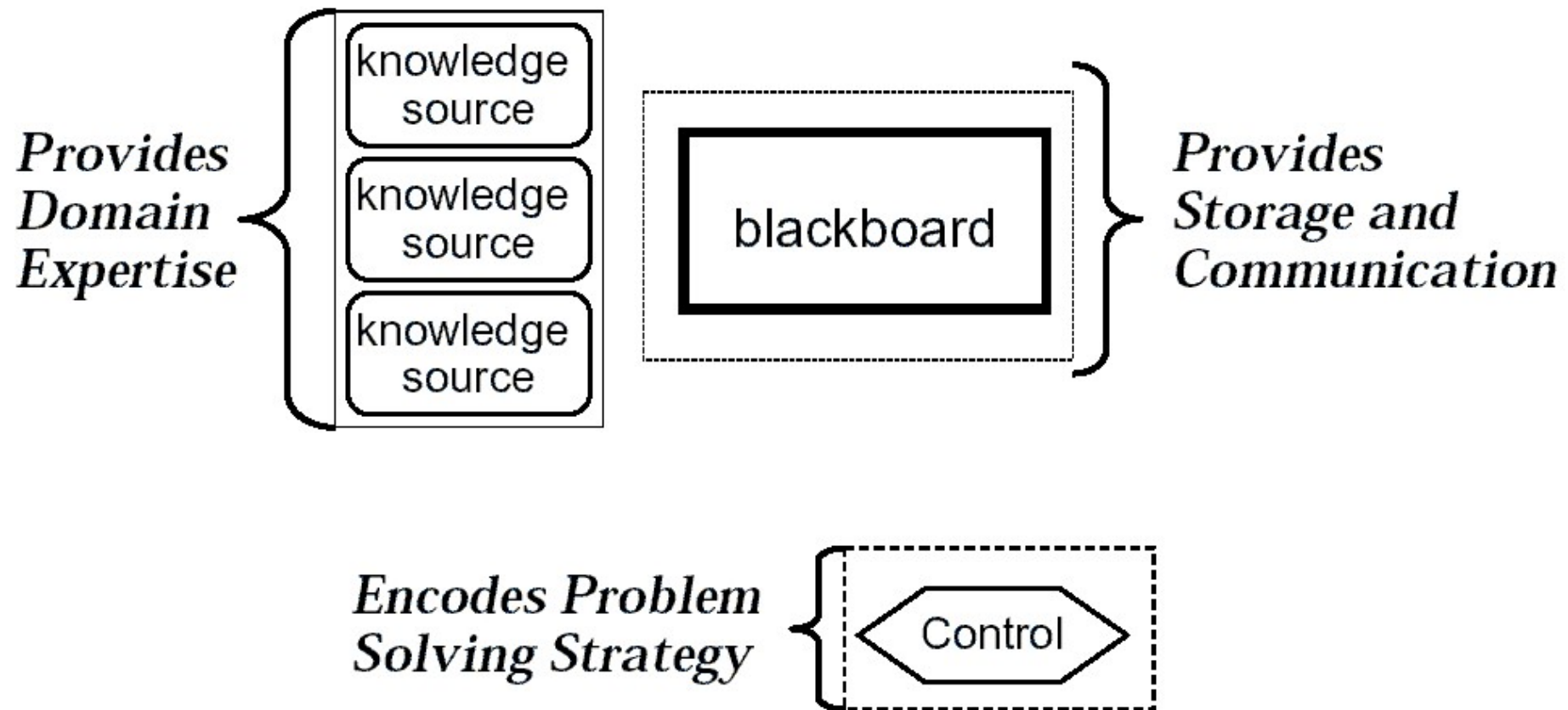


The Blackboard Model



- Knowledge Sources
 - Problem-appropriate knowledge partitioned into separate, independent computations
把问题分成几个部分，每个部分独立计算
 - React to changes in blackboard
响应黑板上的变化
- Blackboard Data Structure
 - Global database containing entire state of problem solution
全局数据库包含解域的全部状态
 - Only means by which knowledge sources interact
知识源互相作用的唯一媒介
- Control
 - Knowledge sources respond
“opportunistically”
让知识源响应偶然事件

The Blackboard Model

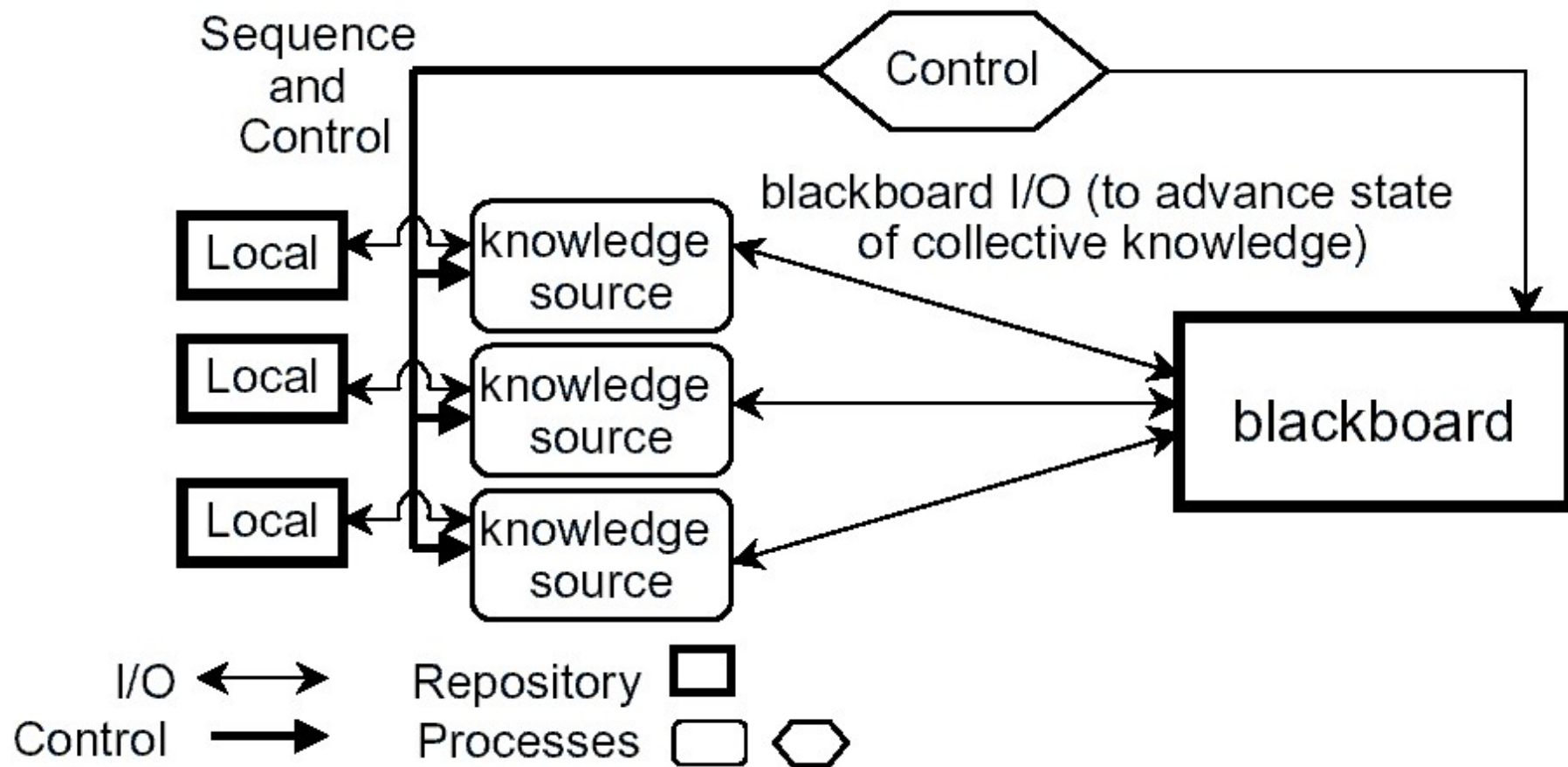


Blackboard Architecture



- Key ideas
 - knowledge shared and distributed among some number of cooperating processes
知识在一些协作的processes之间共享和分布
 - opportunistic processing of data
机会主义地处理数据
- The Blackboard Architecture have 3 major components
 - knowledge sources （知识源）
 - blackboard （黑板）
 - control （控制）
- Historical examples:
 - Hearsay I, Hearsay II, HASP/SAIP, CRYSLIS, ATOME
 - Originally most were from signal processing/interpretation domains
（信号处理、专家系统、模式识别领域经常采用）

Blackboard Architecture





Blackboard Problem Characteristics

- no direct algorithmic solution
没有直接的算法可解
 - multiple approaches to solving the problem
多种方法都可能解决问题
 - various domain expertise required to solve the problem
需要多个领域的专门知识协作解决
- uncertainty
 - error and variability in data and solution
数据和解决方法可能错误或变化
 - moderate to low “signal-to-noise-ratio” in data
数据中信噪比的变化
 - Uncertainty interferes with algorithmic solutions
算法接口的变化



Blackboard Problem Characteristics

- “Best-effort” or approximation is good enough
 - no single discrete answer to problem, or “right” answer may vary
问题没有唯一的解答，或者“正确”答案会变化
- examples
 - signal processing （信号处理）
 - problem solving (planning, logistics, diagnostics)
 - compiler optimization



Knowledge Sources

- Objective 目的:
 - contribute knowledge that leads to solution
提供解决问题的知识
- Representation 表征:
 - procedures, sets of rules, logic assertions
过程、规则、逻辑断言
- Action 功用:
 - modify only the blackboard (or control data -- magic)
只修改黑板
- Responsibility 职责:
 - know when it's possible to help
知道何时能发挥作用
- Selection:
 - loosely-coupled subtasks, or areas of specialization
低耦合的子任务，或者有特别的能力

- Objective 目的:
 - hold data for use by knowledge sources
保存知识源要使用的数据
- Representation 表征:
 - stores objects from solution space, including
保存来自解空间的数据，包括
 - input data, partial solutions, alternatives, final solutions, control data
 - objects and properties define the terms of the discourse
 - relationships are denoted by named links (“next-to”, “part-of”)
- Organization 组织结构:
 - hierarchical, possibly multiple hierarchies; links between objects on same or different levels
分层；链接同层或不同层的对象

- Objective 目的:
 - make knowledge sources respond opportunistically
让知识源响应偶然事件
- Representation 表征:
 - keeps various sorts of information about which knowledge sources could operate and picks a sequence that allows the solution process to proceed a step at a time
了解各个知识源的能力，决策解决问题的步骤
- Remark 备注:
 - the control mechanisms are thoroughly ad hoc
控制机制是彻底的与时俱进

- 优点
 - （1）便于多客户共享大量数据，他们不关心数据何时有、谁提供的、怎样提供的。
 - （2）既便于添加新的作为知识源代理的应用程序，也便于扩展共享的黑板数据结构。
 - （3）知识源可重用。
 - （4）支持容错性和健壮性。

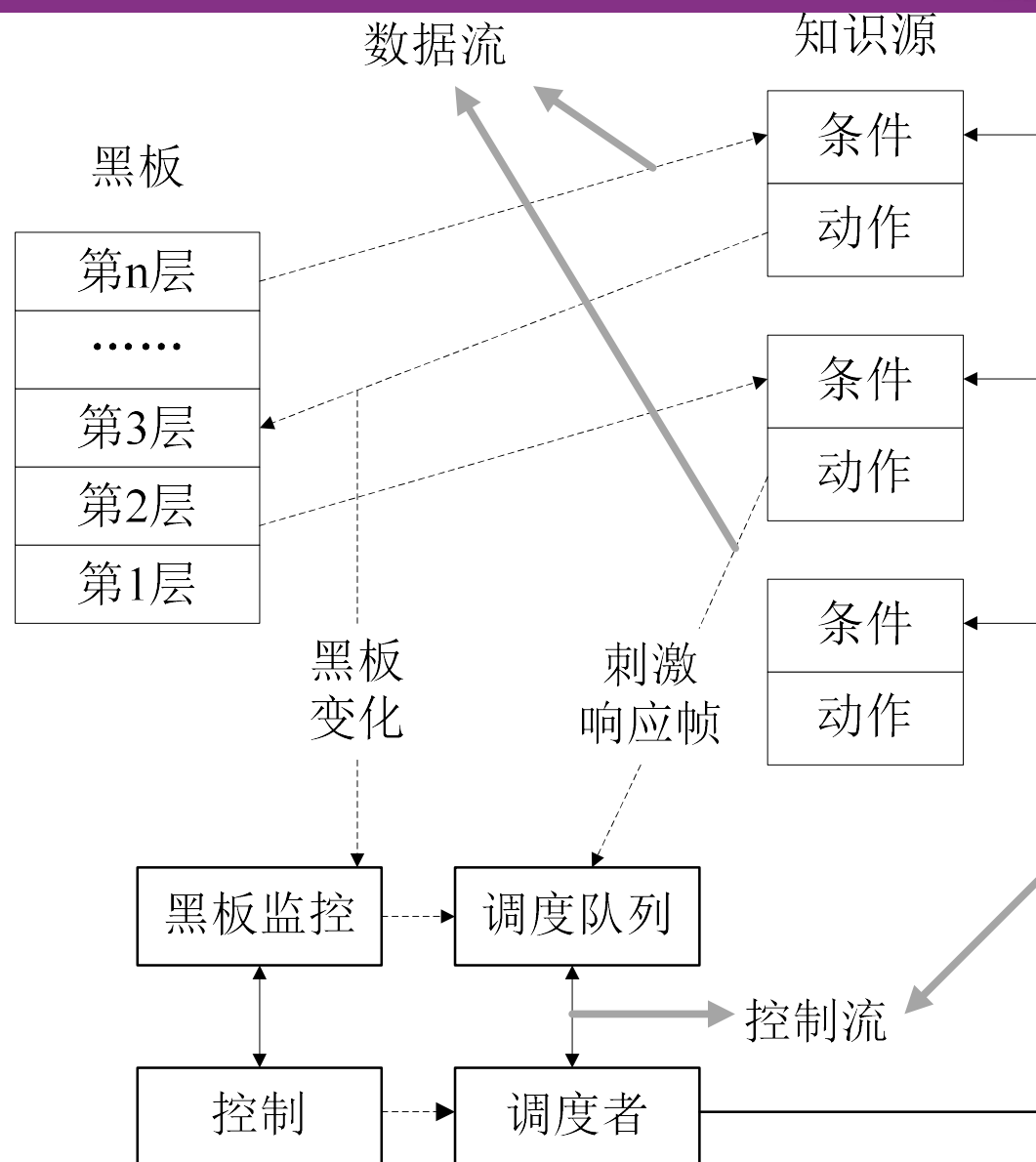
- 缺点
 - 不同的知识源代理对于共享数据结构要达成一致，而且，这也造成对黑板数据结构的修改较为困难——要考虑到各个代理的调用。
 - 需要一定的同步/加锁机制保证数据结构的完整性和一致性，增大了系统复杂度。
 - 不能保证有好的解决方案
 - 测试困难
 - 低效
 - 开发成本高

黑板系统风格



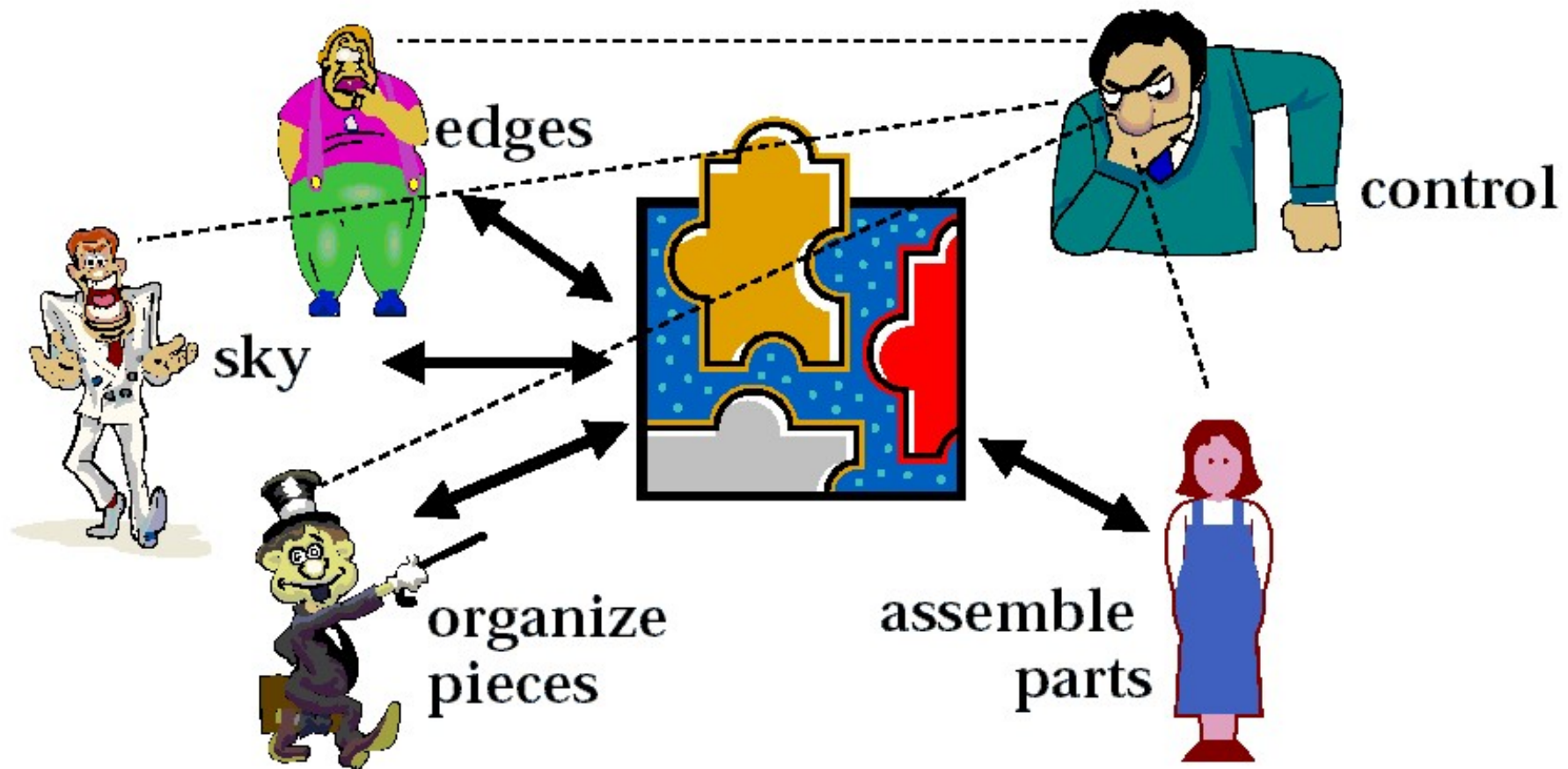
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- 应用实例
 - HEARSAY-II



HEARSAY-II架构

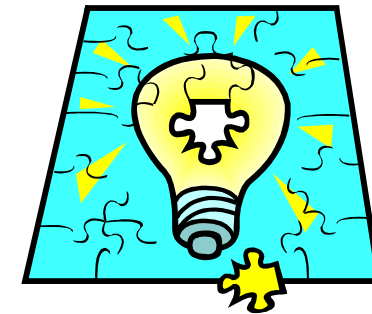
Example: The Puzzle Metaphor



Example: The Puzzle Metaphor

- Data within the blackboard is often structured hierarchically

Level 4	assemble chunks
Level 3	build chunks of edges build chunks of sky
Level 2	collect edge pieces collect sky pieces
Level 1	Turn all pieces picture side up



- Repositories are ubiquitous in computer-based applications.
仓库在计算机应用中几乎无处不在
- Repositories System in many areas show similar trends, stimulated by similar technological and business pressures.
信息共享系统在很多领域有相似的发展趋势，因为有相似的技术和商业压力的激励
- Blackboards are specialized types of repositories for dealing with “messy” information processing domains where the control strategy must be opportunistic.
黑板是一种特别的仓库，主要用于需要对凌乱的信息进行处理的领域，其控制策略必须是机会主义的



Repositories System

- Advantages
 - easy to add consumers and producers of data
很容易增加数据的生产者和消费者
 - How about modifier?
- Issues
 - synchronization (同步)
 - configuration and schema management (配置和管理)
 - atomicity (原子性)
 - consistency (一致性)
 - persistence (持久性)
 - performance (性能)



Repositories System Lineage

- Earliest repositories appear in batch sequential systems
早期的仓库出现在批处理系统
 - mainframes, drums, magnetic tapes, disc drives
 - resources manually managed
人工调配资源
- Pressure for on-line access to data
迫切需要数据即时存取
 - requirement to make access to data easy and instant
简捷地数据存取
 - help to drive the shift from batch-sequential to interactive processing
促进了从批处理到交互式处理的转变



Repositories System Lineage

- Today

- Repositories System appear everywhere from the smallest business, to the most advanced scientific applications

从最小的商业应用到最顶尖的科学应用，信息共享无处不在

- many applications provide access mechanisms to shared data

很多应用提供对共享数据的存取机制

- the Web has become a giant distributed repository
Web成为庞大的分布式仓库